Unit 6, Lesson 15

Base-Ten Blocks to Divide

Standards

Addressing	**4.NBT.B.6**
Building Toward	**4.NBT.A**

Instructional Routines

- MLR7 Compare and Connect
- What Do You Know about ____?

Goals

- Explain (orally) how to use base-ten blocks to represent and calculate the division of a two-digit number by a one-digit divisor.

Student Facing Learning Goals

Let's use base-ten blocks to divide.

Lesson Purpose

The purpose of this lesson is for students to make sense of base-ten representations for division.

Narrative

In the previous lesson, students applied their understanding from IM Grade 3 to divide two- and three-digit numbers by one-digit divisors. Students worked with dividends slightly beyond 100 and represented their thinking in a way that made sense to them.

In this lesson, students work with larger dividends and represent problems with base-ten blocks. This representation emphasizes place value, which supports the work with division in this section. Students are asked to represent their work with base-ten blocks on paper, but that is not the emphasis of this lesson. In the next lesson, students will make sense of and use base-ten diagrams. In future lessons, students will be able to choose a representation and method that makes sense to them as they go deeper into division work.

Access for Students with Disabilities

- Representation

Required Materials

Materials to Gather

- Base-ten blocks: Warm-up, Activity 1, Activity 2
- Tools for creating a display: Activity 1

Lesson Timeline

Warm-up	10 min
Activity 1	25 min
Activity 2	10 min

Teacher Reflection Questions

What surprised you about how students used base-ten blocks to find the value of quotients? How might you use this in tomorrow's lesson?

Synthesis Estimate	10 min
Cool-down	5 min

Warm-up

10 min

What Do You Know about Base-Ten Blocks?

Standards

Building Toward **4.NBT.A**

Instructional Routines

- What Do You Know about ____?

The purpose of this *What Do You Know About ____?* is to invite students to share what they know about base-ten blocks in the context of division. Some students may choose to reflect on base-ten blocks and division, while others may simply describe what they know about base-ten blocks. Any reflection offered by students is useful for activating prior knowledge for the lesson.

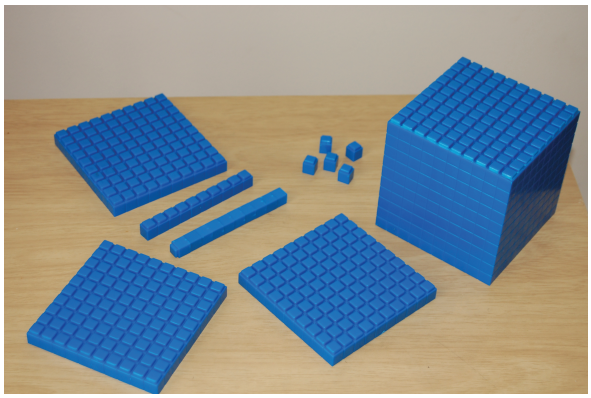
Required Materials

Materials to Gather

- Base-ten blocks: Warm-up

Student Task Statement

What do you know about base-ten blocks?



Student Response

Sample responses:

- They show thousands, hundreds, tens, and ones.
- They can represent different values.
- Each larger block is 10 times the size of the previous block.

Launch

- Display a pile of base-ten blocks.
- "What do you know about base-ten blocks?"
- 1 minute: quiet think time

Activity

- Record responses.

Activity Synthesis

- "Which blocks would we use to represent the number 324?"
- "You have used base-ten blocks to represent large numbers. Today you are going to see how they can be helpful to divide larger numbers."

Activity 1

25 min

Blocks to Divide

Standards

Addressing **4.NBT.B.6**

Instructional Routines

- MLR7 Compare and Connect

The purpose of this activity is for students to find quotients using base-ten blocks and represent their methods. Students solve two problems, one where decomposing a hundred or ten is not necessary and one where it is. Students use base-ten blocks to represent the problem and find the quotient, then they work in small groups to create a visual display of how they used the base-ten blocks. This builds on work students have done in previous grades representing operations with base-ten representations.

The base-ten blocks help highlight the important role place value plays in division (MP7). There are not enough hundreds in 104 to divide into 8 equal groups but there are enough tens to put 1 ten in 8 equal groups. Then the 2 remaining tens can be broken into ones to complete the division.

Required Materials

Materials to Gather

- Base-ten blocks: Activity 1
- Tools for creating a display: Activity 1

Required Preparation

- Each group of 3–4 students needs a set of base-ten blocks that includes 4 hundreds blocks, 10 ten blocks, and 25 ones blocks.

Student Task Statement

Use the base-ten blocks to represent each expression. Then find the value of each expression.

1. $488 \div 4$
2. $104 \div 8$

Student Response

1. 122. Sample response: Students split the 4 hundreds into 4 groups and then do the same with the 8 tens and 8 ones. They see they have 1 hundred, 2 tens, and 2 ones in each group.
2. 13. Sample response: Students trade the hundred block in for 10 tens. They split the 10 tens into 8 groups and are left with 2 tens. They split the 2 tens into 20 ones and then split the 24 ones into 8 groups.

Launch

- Groups of 3–4
- Give each group at least 4 hundreds blocks, 10 tens blocks, and 25 ones blocks.
- "Work with your group to represent each expression with base-ten blocks, then find the quotient."

Activity

- 5–6 minutes: group work time
- "How was finding each quotient the same? How was it different?" (With the first one, you could just split the blocks into 4 groups. With the second one, you had to trade in some blocks to make 8 groups.)

MLR7 Compare and Connect

- Give each group tools for making a visual display.
- "Create a visual display that shows how you used the

base-ten blocks to find the quotient, including details such as notes, diagrams, drawings, and so on, to help others understand your thinking."

- 2–5 minutes: group work
- 5–7 minutes: gallery walk
- "As you look at the displays from other groups, record things that are the same and things that are different."

Activity Synthesis

- "What is the same and what is different between the representations?"
- 30 seconds quiet think time
- 1 minute: partner discussion

Advancing Student Thinking

If students find the value of each expression using only numbers and symbols, consider asking:

- "How did you find the value of this expression?"
- "How can you use base-ten blocks to represent the division and show your way of thinking about it?"

Activity 2

10 min

Show Us Your Blocks

Standards

Addressing **4.NBT.B.6**

The purpose of this activity is for students to find quotients and represent their thinking with base-ten blocks. The numbers in the expressions are designed to encourage students to think about when the base-ten blocks may be helpful and when they become cumbersome. In future lessons, students will be asked to interpret base-ten representations but may use any method or representation that makes sense to them.

Access for Students with Disabilities

Representation: Develop Language and Symbols. Synthesis: Use gestures or annotations such as labels or arrows to make connections between representations on displayed student work.
Supports accessibility for: Conceptual Processing, Visual-Spatial Processing

Required Materials

Materials to Gather

- Base-ten blocks: Activity 2

Student Task Statement

Find the value of each expression. Explain or show your reasoning. Use base-ten blocks if it helps.

1. $96 \div 4$
2. $86 \div 2$
3. $108 \div 9$

Student Response

1. 24. Sample response: Draw 4 groups with 2 tens in each group. Divide remaining 16 and draw 4 ones in each group.
2. 43. Sample response: Draw 2 groups with 4 tens in each group. Draw 3 ones in each group.
3. 12. Sample response: Start with 10 tens and draw 1 ten in each of 9 groups. Divide remaining 18 and draw 2 ones in each group.

Launch

- Groups of 2
- Give students access to base-ten blocks.
- "You are going to solve some problems on your own. Use base-ten blocks and represent your thinking in your book."

Activity

- 5–6 minutes: independent work time
- 2 minutes: partner discussion
- Monitor for and select students who draw base-ten representations using small and large squares and rectangles labeled with numbers.

Activity Synthesis

- "In the *Lesson Synthesis*, let's compare some different representations for a division problem."

Advancing Student Thinking

If students use mental math to solve problems, consider asking:

- "Tell me more about how you found the value of _____."
- "How might you use the blocks to check your mental math?"

Lesson Synthesis

Display student work showing base-ten representations for the last expression ($108 \div 9$).

"How did this student represent their reasoning?" (They drew 10 tens and then crossed out 2 tens and drew 20.)

"Do you have any questions or suggestions that could help them make their work clearer?" (They could add numbers to label the parts.)

Repeat with other student's work, as time allows.

"Take a minute, and if you'd like, revise your work to make it easier for someone else to understand."

Suggested Centers

- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)
- Rolling for Fractions (3–5), Stage 2: Multiply a Fraction by a Whole Number (Supporting)

Cool-down

5 min

Division Reflection

Standards

Addressing **4.NBT.B.6**

Student Task Statement

How was using the base-ten blocks helpful in your work today? How was it not helpful?

Student Response

Sample response: It was helpful when we were working with smaller numbers and we didn't have to decompose blocks. It wasn't helpful when I was trying to work with larger numbers.

Responding to Student Thinking

This lesson builds on concepts of division developed in a prior unit.

Prior-Unit Support
Grade 3, Unit 4, Section D Dividing Greater Numbers